Geometric Coordination and Residual Reinforcement Learning for Robust Bimanual Handover

Project goal. I plan to investigate geometric coordination and lightweight residual reinforcement learning for bimanual object handover in simulation. One arm holds a rigid object while another grasps it and takes over support. Spatial errors can leave the receiving gripper misaligned with the object; timing errors can cause release before a stable receiving grasp is established. I will test whether combining explicit coordination with learned position corrections improves handover success under these perturbations compared with either method alone.

Prior work has studied reinforcement learning for dexterous object handover, including robustness to unseen objects and motion perturbations [1]. ManipTrans uses residual learning under interaction constraints to transfer human bimanual demonstrations to robotic hands [2]. These studies motivate my controlled comparison of coordination and residual learning using parallel-jaw grippers.

Proposed approach. I will build a handover environment in Isaac Sim using Isaac Lab, with two parallel-jaw grippers and low-dimensional state observations. A scripted controller will provide nominal trajectories and gripper commands. A geometric coordination module will adjust relative gripper targets and use contact and motion observations to defer release until the receiving gripper has secured the object. This module will require no policy training.

I will train a small residual policy using proximal policy optimization (PPO) to make bounded position corrections to the fixed nominal controller. The policy will use relative poses, motion information, and the contact phase. In the combined method, the geometric module will process these corrections and govern release timing through contact conditions. The residual policy will adjust positions, while orientation targets remain fixed.

Evaluation. I will compare the nominal controller with geometric coordination, residual learning, both, or neither. Without geometric coordination, nominal release commands remain unchanged. An ablation will remove the combined policy's contact phase input. Experiments will perturb initial object pose and nominal release timing, separately and jointly, with physical parameters fixed. Under identical evaluation conditions and matched training budgets, I will report success and drop rates, learning curves, completion time, and collisions. Success requires independent holding by the receiving arm for two seconds after release. Failure analysis will distinguish misalignment, premature release, and conflicting motions during shared holding.

Scope and expected outcome. I will focus on one simulated handover task, first establishing the baseline, then implementing the modules and conducting comparisons. Deliverables will include reproducible code, quantitative results, and demonstration videos documenting the benefits and limitations of combining the methods.

Disclosures. I will work independently, with no collaborators inside or outside the class. This project will not be used for another course and is not part of my ongoing research. Work so far is limited to topic selection and proposal planning; no implementation or experiments have been completed.

References

[1] D. Frau-Alfaro, J. Castaño-Amoros, S. Puente, P. Gil, and R. Calandra. Learning Dexterous Object Handover. RO-MAN, 2025.

[2] K. Li, P. Li, T. Liu, Y. Li, and S. Huang. ManipTrans: Efficient Dexterous Bimanual Manipulation Transfer via Residual Learning. CVPR, 2025.